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Magnetic and Paleomagnetic Studies of the Nipissing Diabase, Lake Matinenda Area, Ontario

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Samples of the Nipissing diabase were collected for rock-magnetic and paleomagnetic studies around and west of Lake Matinenda, Ontario. The specimens have been classified into three groups, based on their stability of natural remanent magnetism, using the alternating field demagnetization technique.

Opaque petrological studies showed that the specimens which possessed magnetite–ilmenite exsolution were magnetically very stable. The specimens without any exsolution possessed weak and unstable magnetization.

Sphene was identified in the alteration product of ilmenite using X-ray techniques. Curie point measurements indicated low titanium content in the specimens which contained exsolution of ilmenite in magnetite.

The pole position, calculated from five significant sites located around and west of Lake Matinenda is 69.9° W, 15.6° S ($A_{95} = 6.1°$).

Des échantillons du diabase de Nipissing ont été collectionnés afin d'étudier le paléomagnétisme et le magnétisme des roches situées à l'ouest et autour du Lac Matinenda, Ontario. Les spécimens furent classifiés en trois groupes, basés sur la stabilité du magnétisme rémanant naturel déterminé par la technique de démagnétisation du champ alternatif.

Des études pétrologiques ont montré que les échantillons possédant des exsolutions ilménite-magnétite sont magnétiquement très stables, tandis que les spécimens sans exsolution ne présentent qu'une magnétisation faible et instable.

Des analyses aux rayons-X ont permi de déterminer la présence de sphène dans le produit d'altération. Des mesures du point Curie ont conduit à la détection d'un faible contenu en titane dans les échantillons qui contenaient des exsolutions d'ilménite dans la magnétite.

La position des pôles, calculée à partir de cinq points significatifs situés autour et à l'ouest du Lac Matinenda est 69.9° W, 15.6° S ($A_{95} = 6.1°$). [Traduit par le journal]

Introduction

The Nipissing Diabase outcrops over a wide area of the Northern shore of Lake Huron and in the Cobalt region in the Province of Ontario. The diabase is intruded either as near vertical dikes or as flat-lying to gently undulating sill-like bodies into the Huronian sedimentary rocks. The diabase of the Lake Matinenda area, north of Blind River, shows no clear cut relationship between the sills and the dikes. The dikes were probably intruded at much the same time as sills or they may have acted as feeders to the sills from the same magma (Robertson 1963).

The Nipissing diabase from the Cobalt area

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has been dated at 2095 m.y. by K-Ar in biotite (Lowdon *et al.* 1963). The Rb-Sr age of the diabase from the Blind River area was given as 2155 ± 80 m.y., based on whole-rock isochrons (Van Schmus 1965). In many areas, the sills show development of a granophyre or the 'red-rock' usually at the top due to differentiation in larger bodies (Robertson 1962). The texture of this rock is medium to coarse and red colour is imparted by hematite in feldspar. This granophyre was dated at 1700 ± 50 m.y. on mineral separates (feldspar) by Van Schumus (1965). This date is interpreted as a metamorphic event during the Hudsonian Orogeny.

Earlier paleomagnetic studies on the Cobalt area diabase have been carried out by two authors. Strangway (1964) studied the intensity and the direction of the remanent magnetism of the diabase sampled near Cobalt township, to estimate the age of the diabase on the basis of the rate of decay of the stable remanence intensity. Symons (1967) collected thirteen samples of the diabase, along with other Precambrian rock samples, from the Silverfields Mines near Cobalt, and showed that the paleomagnetic results were consistent with the sequence of geologic events for the Cobalt area. In 1970, Symons collected samples from fourteen additional sites from the Cobalt–Silver area. The remanence directions of these samples were analysed to prove that the diabase acquired its remanence after the formation of the Kerr Lake Arch and the North Lorrain Basin structures. In 1971, Symons sampled 19 sites (sills and dikes) of the same rock in the Blind River–Elliot Lake area of Ontario. From the results of paleomagnetic studies of the diabase, he tentatively concluded that the sheets and dikes of this area contained generally weak but stable primary remanence components which tend to be masked by much stronger unstable components. He also concluded that the sheets were emplaced and cooled during several polarity intervals of the earth's magnetic field about 2155 m.y. ago.

Sampling and Laboratory Procedures

Eleven sites of the diabase sills were sampled using a portable diamond drill, taking four to five cores from each site. These sites are located around and west of Lake Matinenda as shown in Fig. 1, and include three sites of the granophyre and eight of the normal diabase. Sites M2A, M3A, M6A, M8A, and M11A of the diabase were sampled on the shores of the Lake. Site M36A of the diabase was sampled 36 mi (58 km) from Elliot Lake on Highway 639. The Nipissing granophyre was sampled at three locations. The samples of Site M34A were collected from the ferry landing to St. Joseph's Island 1.5 mi (2.4 km) south of Highway 17. Site M35A was sampled from road-cut exposures 11.1 mi (17.9 km) north of Highway 17 on Highway 129. The last site of the granophyre, site M37A, was located on a northwest tract 11.5 mi (18 km) away from a northerly turnoff from Highway 17, 2.5 mi (4 km) west of Blind River.

The field-drilled core samples were oriented *in situ* by a device similar to that described by Helsley (1967). The bearing of the arrow, marked on the core, was read first using a magnetic compass and was checked using a modified Mark II astro-compass. If the difference between the readings of the sun compass and the magnetic compass was less than 3°, then an average value of the two readings was taken as the azimuth of the arrow marked. At a few sites, the rock was strongly magnetized. Consequently the magnetic compass readings were appreciably distorted. In such cases, only the sun compass readings were relied upon.

Direction and intensity of natural remanent magnetization of each of the specimens was first measured using a commercial spinner magnetometer (Princeton Applied Research-Model SM-1). Any viscous secondary components of magnetization present were progressively demagnetized using an alternating field technique, on a three axis tumbler apparatus. The field which gave the best precision within a site was defined as the 'Optimum field' for that particular site. The maximum field used for the treatment was 1200 Oe peak. One polished section was also prepared from each core sample to study opaque grains under the microscope. The measurements of the Curie points were carried out on a Curie balance similar to that of Chevallier and Pierre (1932) with a bifilar suspension. The field in which the measurements were made was 1640 Oe.

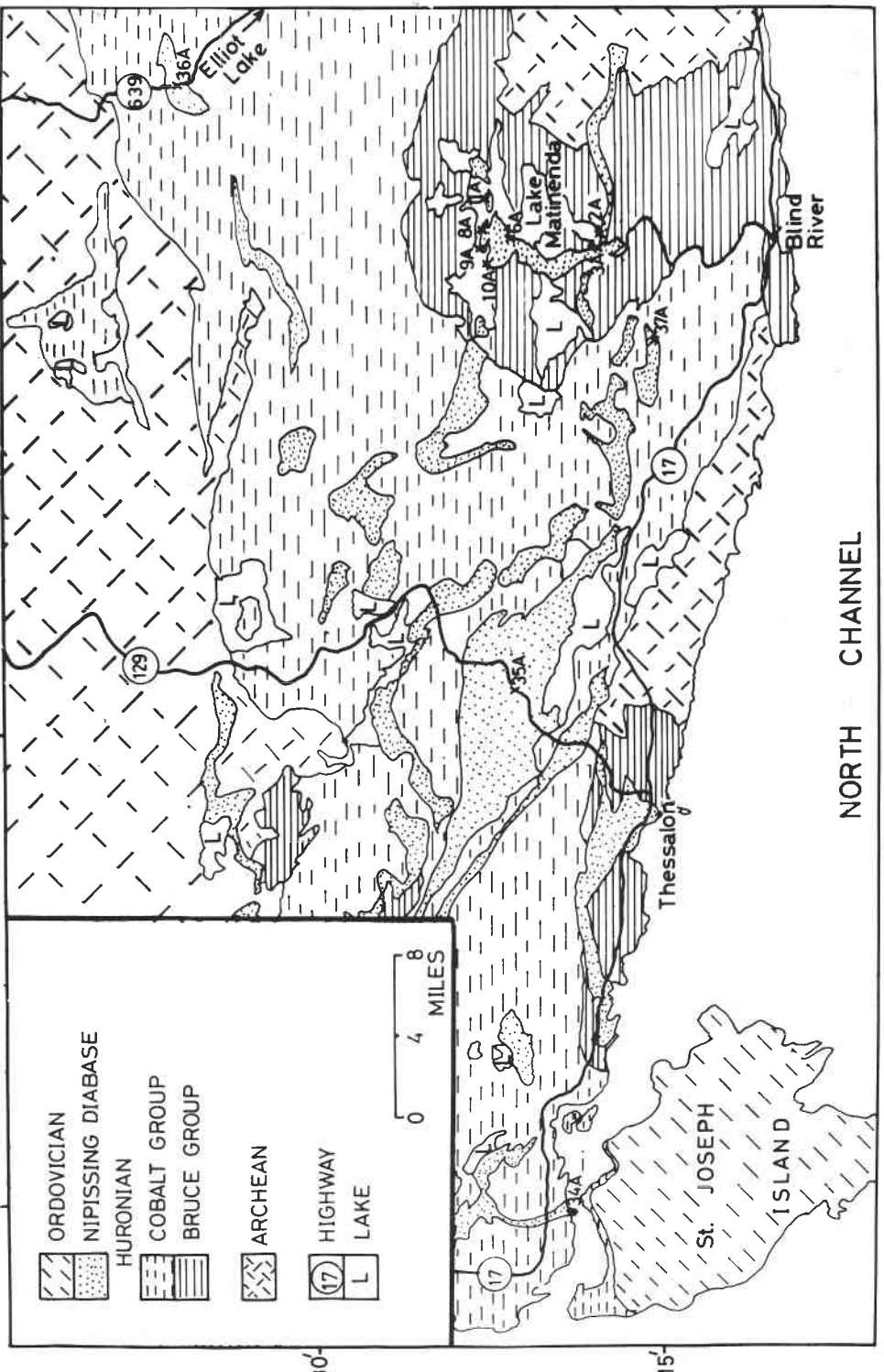


FIG. 1. Generalized geological map showing sampling sites.

Results and Discussion

Site mean directions of the eleven sites before and after demagnetization are shown in Fig. 2. The inclinations of Sites M34A, M35A, and M37A (Nipissing granophyre) point downward and their directions lie away from the general clustering area of the diabase sites. The inclinations of site M10A and M11A (Nipissing diabase) also point downward, though their azimuths tend to group with other diabase site mean directions after 'cleaning'.

The variations in within-site dispersion may be illustrated by plotting k/k_{sig} vs. the strength of the applied alternating field (Irving *et al.* 1961). The parameter k is the best estimate of the dispersion parameter κ (Fisher 1953) and k-significant (k_{sig}) is equal to $(N-1)/(N-Ro)$ where Ro is the value, tabulated by Watson (1956*b*), of the resultant of N random vectors which will be exceeded with a probability of 5%. If R, the resultant vector of samples of size N, is less than or equal to Ro, $k/k_{sig} \leqslant 1$, and the distribution may be considered random. When R exceeds Ro, k/k_{sig} is larger than unity and its value is a measure of the tightness of grouping. The eleven sites have been classified into three groups based on the above criterion.

Group A

The specimens from sites M2A, M3A, and M6A were extremely stable magnetically. These sites, illustrated in Fig. 3, showed poor precision within the sites, initially. With higher fields of demagnetization, the within-site precision improved markedly. The best grouping of the directions was obtained around 1000 Oe peak field. The characteristic A.F. demagnetization curves for the typical specimens of the above sites are also shown in Fig. 4. The specimens of these sites are magnetically hard and possess high coercivity of remanence.

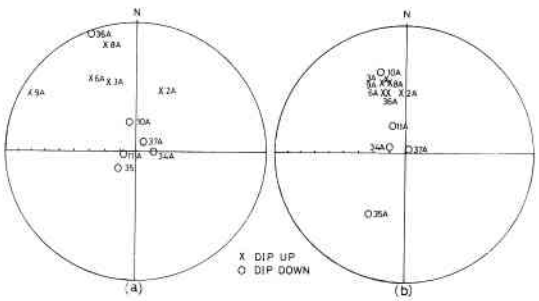


FIG. 2. The site mean remanence directions (*a*) before 'cleaning' (*b*) after 'cleaning'.

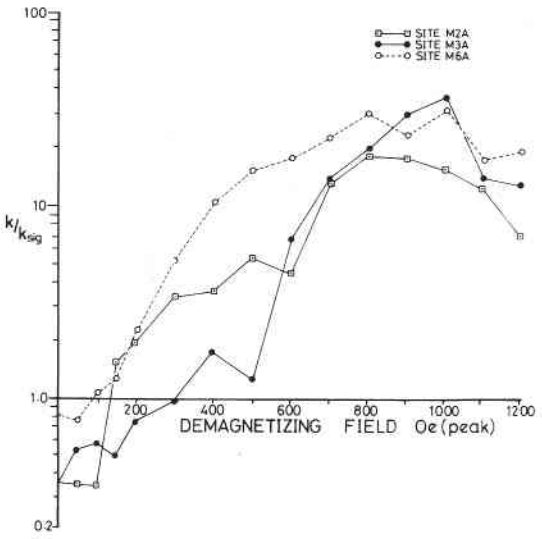


FIG. 3. The within-site dispersion for Group A sites, illustrated by plotting paramater k/k_{sig} against the demagnetizing field.

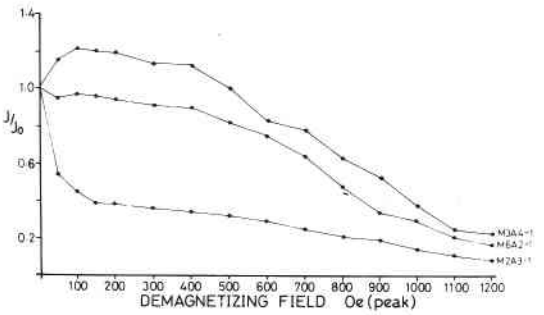


FIG. 4. Normalized A.F. demagnetization curves for Group A specimens.

Group B

The k/k_{sig} ratios of sites M8A, M35A, M36A, and M37A oscillate with higher fields of demagnetization as shown in Fig. 5. Although site M35A is nonrandom according to the criterion used, its k/k_{sig} ratio remained very close to unity. The typical A.F. demagnetizing curves, shown in Fig. 6, indicate that about 80% of the intensity of the natural remanence is lost after 100–150 Oe (peak) field treatment.

Group C

The sites M9A, M10A, M11A, and M34A

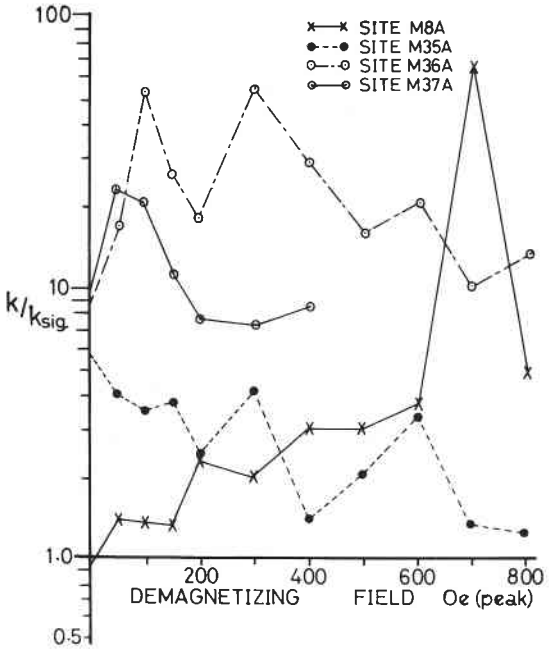


FIG. 5. The within-site dispersion for the Group B sites.

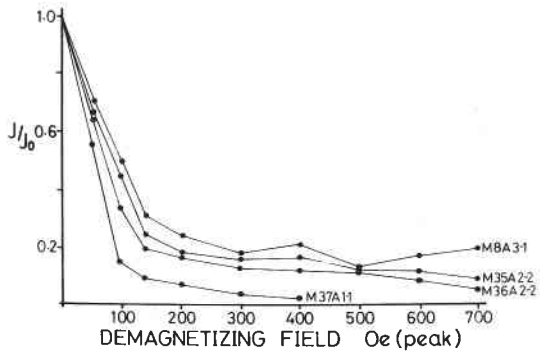


FIG. 6. Typical normalized A.F. demagnetization curves of the specimens from Group B sites.

were considered random. The k/k_{sig} ratios against the alternating field and the typical A.F. demagnetization curves are illustrated in Figs. 7 and 8, respectively. About 90% of the intensity of the natural remanence is lost after 100–150 Oe peak field. The directions remained inconsistent throughout the process of demagnetization. The intensity of natural remanence became too weak to be measured after 800 Oe field.

Polished Section Petrology

The characteristic alternating field demagnetization curves of the above three groups of specimens indicated a wide coercive force spectrum of the grains carrying the magnetization. An effort was made to correlate the polished section petrology with the stability of remanence of the specimens of the above three groups.

Distinct exsolutions were observed in the Group A specimens under reflected light. First, a well developed structure of discontinuous and parallel hematite lamellae was observed in ilmenite grains as shown in Plate 1*a*. All the host grains were completely filled with such fine hematite lamellae, roughly 4–8 μ long and 0.5 μ wide. Secondly, ilmenite lamellae were observed in magnetite grains. The ilmenite lamellae occur in two different sets. The first set of lamellae about 2–3 μ broad, run right across the magnetite grains (Plate 1*b*). The

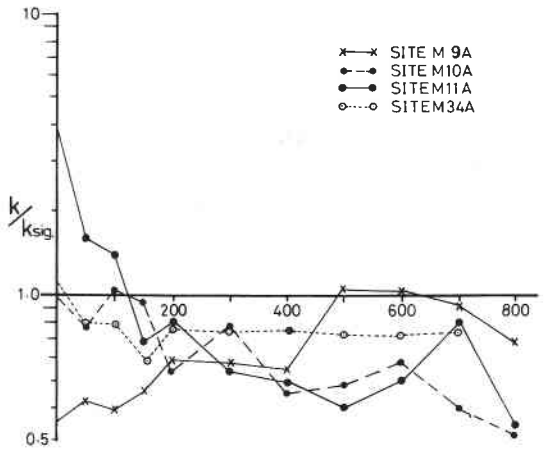


FIG. 7. This figure illustrates the k/k_{sig} ratio against the demagnetizing field (Oe peak) for the Group C sites.

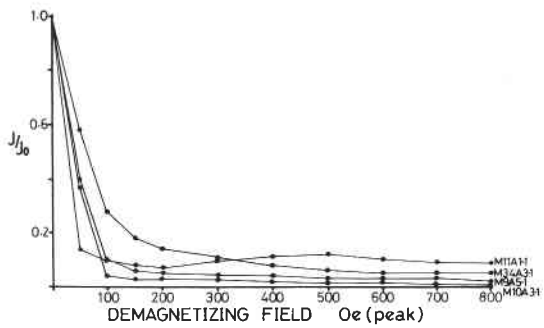


FIG. 8. Typical normalized A.F. demagnetization curves for the Group C specimens.

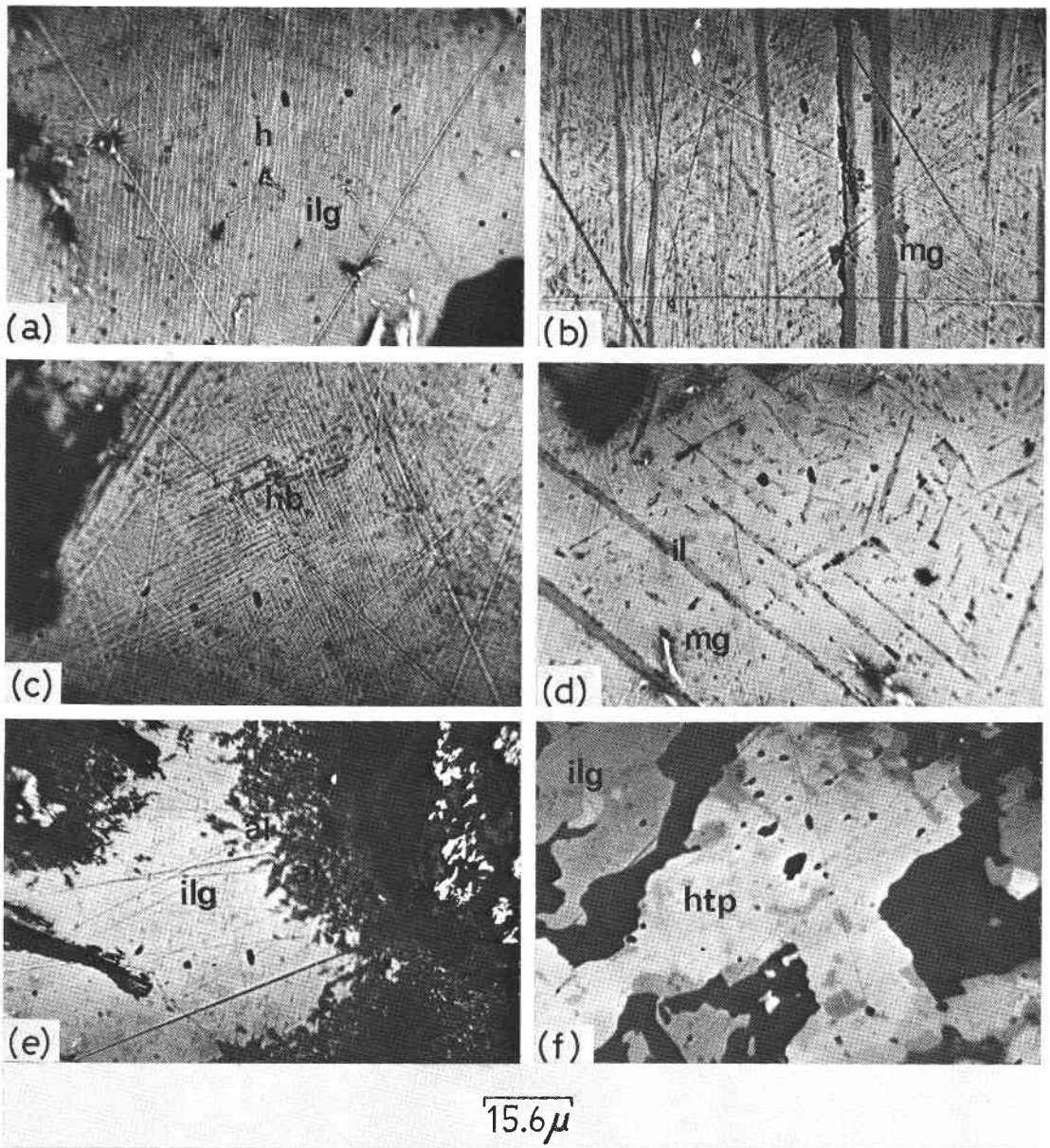


Plate 1

(*a*) Discontinuous hematite lamellae (h) in ilmenite grain (ilg).
(*b*) Broad ilmenite lamellae (il) in magnetite grain (mg).
(*c*) 'Herring-bone' structure (hb) of ilmenite lamellae in magnetite grain.
(*d*) Magnetite grain (mg) partly divided by the broad ilmenite lamellae (il).
(*e*) Alteration product (al) in ilmenite grain (ilg).
(*f*) Suspected hematite phase (htp) in ilmenite grains (ilg).
(All photomicrographs were taken using an oil immersion objective)

second pattern of the ilmenite lamellae exhibits 'Herring-bone' type of structures as shown in Plate 1*c*. In each polished section of this group, the ilmenite–hematite exsolution dominated the magnetite–ilmenite exsolution. The stable natural remanence and the coercive force greater than 1200 Oe observed in specimens of this group may be due to both the exsolved phases. The magnetic grains are possibly broken into single domains by the fine herring-bone structure of ilmenite (Evans and McElhinny 1966). The fine elongated hematite lamellae in ilmenite grains may also have contributed to the high coercive remanence and the stable magnetism of the specimens (Carmichael 1961).

In the case of Group B specimens, the magnetite grains were partly filled with broad ilmenite lamellae, about 25–30 μ long and 2 μ wide (Plate 1*d*). Homogeneous magnetite grains were few. This exsolution possibly belongs to Class II of titanomagnetite high temperature oxidation described by Wilson and Haggerty (1966) and Ade Hall *et al.* (1968). The natural remanence is carried by the magnetite grains partly divided by the broad ilmenite lamellae. No other exsolved phases were observed in specimens of this group.

Altered skeletal ilmenite and large angular pyrite grains were the only grains observed in Group C specimens. The alteration in ilmenite occurred on the periphery of the grains as shown in Plate 1*e*. The altered product of the ilmenite was suspected to carry a few minute grains of ferromagnetic material which might have been responsible for carrying the weak remanence. The altered products were analysed using an X-ray powder technique (Debye-Scherrer, 57.26 mm diameter camera). The procedure for the preparation of the specimen for this camera has been outlined by Patel (1971). Strong lines of sphene, ilmenite, augite, and plagioclase were observed on the films. No trace of ferromagnetic material was observed in the altered product of the ilmenite.

Pale yellow to whitish anisotropic irregular patches (Plate 1*f*) were also observed in some of the ilmenite grains. This may be initial development of a hematite phase and is perhaps responsible for carrying the weak and unstable magnetization.

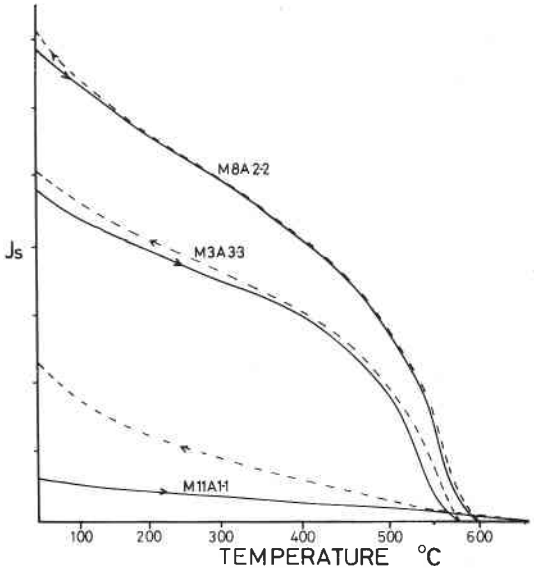


FIG. 9. Typical thermomagnetic curves (*Js* vs. *T*). *Js* is the saturation magnetization in arbitrary units. Solid lines are heating curves, dashed lines cooling curves.

Curie Point Analysis

The typical saturation magnetization (*Js*) versus temperature curves for the three groups of specimens are shown in Fig. 9. The Curie point obtained for the Group A specimens (M3A3-3) was about 555 ± 5 °C. It appears from this Curie temperature observation that the material of this Group is mainly magnetite. The ilmenite–hematite phase could give the same Curie temperature if it has a suitable composition, and if so, the self-reversing mechanism in ilmenite–hematite solid solution is ruled out (Uyeda 1958; Carmichael 1961). But it is possible that this solution will not show up on the *Js*/*T* curves, as it would be masked by magnetically stronger magnetite–ilmenite exsolution.

The Group B specimens (M8A 2-2), with ilmenite lamellae exsolved in magnetite grains showed a single Curie point of about 570 ± 5 °C. It appears from this Curie point that the magnetite in these specimens has a low titanium content as the Curie point of pure magnetite is 580 °C. The third type of thermomagnetic curve is shown by the specimen M11A1-1 Group C category, which is extremely unstable magnetically. The curves have long 'tails' and

TABLE 1. Pole from mean direction of significant sites

Site number	N	k	α°_{95}	Mean dec.	Mean inc.	Pole positions			
						Lat. °(N)	Long. °(W)	δp	δm
M2A	4	59.9	11.7	355.3	−50.2	−12.5	78.9	10.5	15.8
M3A	5	96.1	8.3	344.7	−38.8	−20.3	67.9	7.5	18.3
M6A	4	105.5	9.0	339.6	−47.1	−13.2	64.8	8.0	11.6
M8A	4	9.9	6.2	346.5	−41.1	−19.0	70.1	4.8	7.7
M35A	4	10.6	25.5	211.0	44.6	−21.1	115.5	21.7	32.6
M36A	4	52.8	6.9	343.4	−48.3	−12.8	67.6	6.2	8.9
M37A	5	19.6	15.2	78.4	84.9	47.3	68.7	30.0	33. 0

Mean of poles of significant sites (excluding sites M35A and M37A): 69.9° W, 15.6° S (A°_{95} = 6.1°).
N = Number of Samples.
k = Estimate of Fisher's precision parameter.
Dec. = Declination in degrees east of true north.
Inc. = Inclination in degrees measured positive downward.
α°_{95}, A°_{95} = Radius of circle of 95% confidence.
δp, δm = Semiaxes of 95% confidence oval.

no well defined Curie points. Similar behavior was also observed in basic lavas by Ade Hall *et al.* (1965) and in Tertiary volcanic rocks by Raja (1968). The saturation magnetization moments are low which agrees with the observed polished section petrology.

Pole Position

The pole positions of the seven significant sites (Groups A and B) of the diabase are given in Table 1. The mean of the poles of the five sites (M2A, M3A, M6A, M8A, and M36A) is 69.9° W, 15.6° S (A_{95} = 6.1°). Sites M35A and M37A (granophyre) were discarded for calculating the mean pole as the development of the granophyre may be a metamorphic event and their magnetic directions diverge by more than 90° from the directions of the other five sites.

The pole positions of the same formation in other parts of the Province have also been determined by Patel (1971). The mean of 27 sites in the Cobalt area is 95.0° W, 10.1° S and Symons (1970) calculated a pole at 88.1° W, 19.4° S from 11 sites. Symons (1971) also determined the pole position for the Elliot Lake area at 98.4° W, 0.8° S from eight sites. The comparison of mean directions was carried out with Patel's (1971) results to that of Symons' (1970) results and Symons' (1971), results to that of this paper. In both the cases, the directions are observed nonsignificantly different using Watson's (1956*a*) F-ratio test at the 5% significance point.

Conclusion

The correlation of the opaque petrology with the stability of remanence of the three groups of specimens appears to be significant.

Sites M9A, M10A, M11A, and M34A showed random distribution using the Watson's (1956*b*) test. The stability of remanence of the individual specimens from these sites was in keeping with the opaque petrology. Sites M2A, M3A, and M6A were found to be extremely stable. The specimens from these sites contained two exsolutions. The varied differences in high temperature oxidation of the minerals of the specimens from the same area may be due to different oxidation environments to which the sills have been subjected.

It is rather difficult to comment on the titanium contents from the Curie points of specimens of Group A as both the exsolutions are present in the same specimens. The magnetite in Group B specimens has little or no titanium and has a Curie temperature of 570° C.

The mean pole position for the Nipissing diabase around and west of Lake Matinenda is calculated at 69.9° W, 15.6° S (A_{95} = 6.1°), using the pole positions of five significant sites.

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